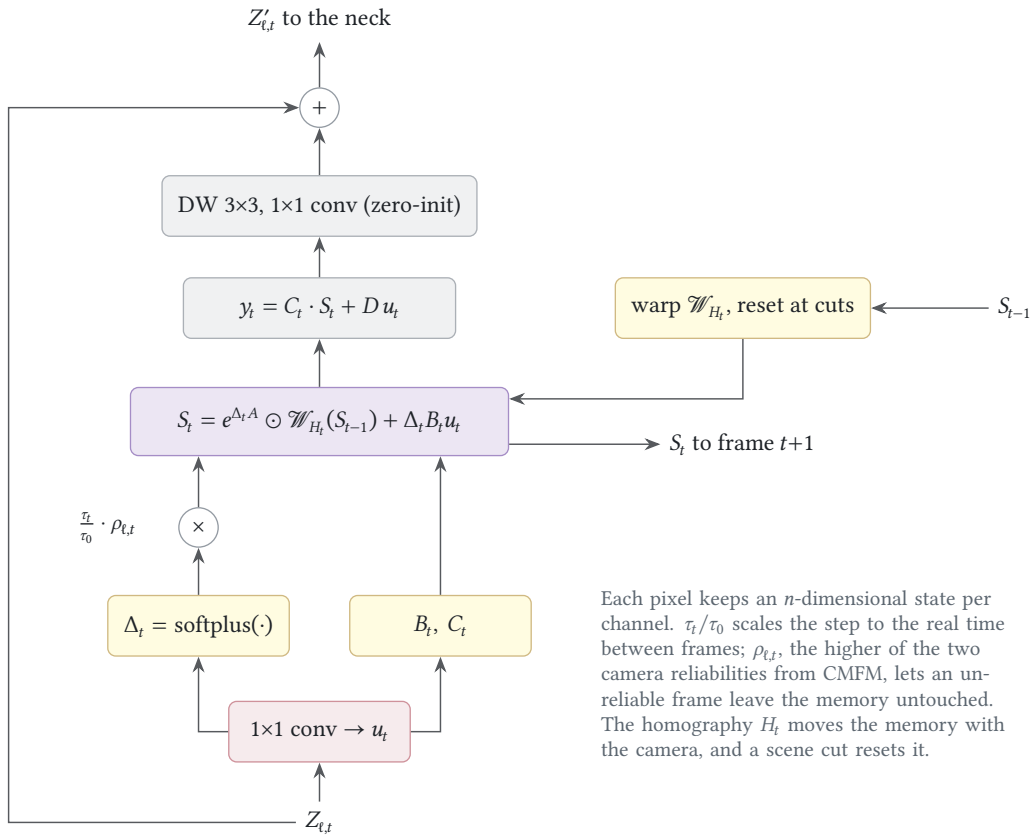
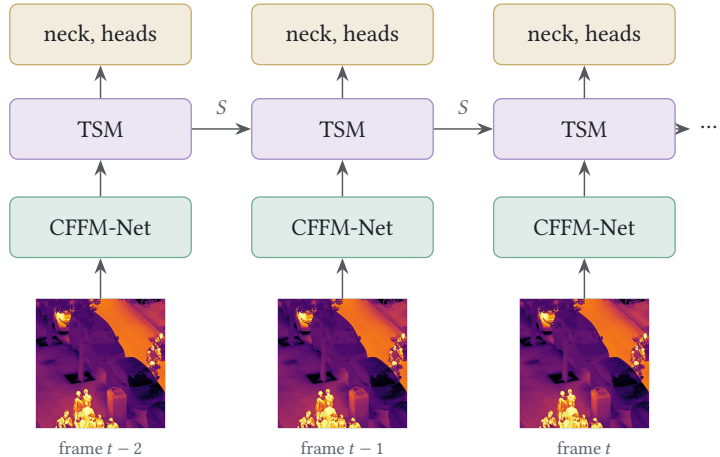
(a) one frame at one pyramid level



Each pixel keeps an n-dimensional state per channel. τ_t/τ_0 scales the step to the real time between frames; $\rho_{\ell,t}$, the higher of the two camera reliabilities from CMFM, lets an unreliable frame leave the memory untouched. The homography H_t moves the memory with the camera, and a scene cut resets it.

(b) unrolled over a stream



The fused features of each frame update a per-pixel memory, so a person who drops below the threshold in one frame is still remembered in the next. Phase 2 trains this on paired visible–thermal video.